

Towards Interpretable Twin Support Vector Machines with Probabilistic Class Membership

Revision on TWSVM

Compared to the standard linear Support Vector Machine (SVM) model, Twin Support Vector Machine (TWSVM) solves two small quadratic programming problems (QPPs) instead of a large one. This ensures TWSVM is competitive in terms of computational time being four times faster than SVM as well as even more effective than SVM on several benchmark datasets specially when dealing with specific datasets such as the "Cross Planes" datasets.

Given a typical binary classification problem with M observations over N features, $X \in R^{M \times N}$, such that an observations might belong to one of two possible classes, either class 1, $x_m = +1$ or class -1 $x_m = -1$.

Assume all the observations belonging to class 1 are included in a matrix $A \in R^{m_1 \times N}$ where m_1 is the number of observations of class 1. And $B \in R^{m_2 \times N}$ is a matrix including all observations on class -1.

In this context, the TWSVM algorithm seeks two non-parallel hyperplanes in the n -dimensional input space, namely, $x^T w_1 + b_1 = 0$ and $x^T w_2 + b_2 = 0$ such that, each of these hyperplanes is as close as possible to the samples of its own class and as far as possible from the samples of the other class.

Demanding the hyperplanes to be as close as possible to the samples of its own class is equivalent to minimize the sum of squared scaled distances for all the observations of a given class with respect to their corresponding hyperplane. This leads to solving the following Quadratic Programming Problems (QPPs),

$$\min_{w_1, b_1} \sum_{i=1}^{m_1} (x_i^T w_1 + b_1)^2 = \min_{w_1, b_1} \frac{1}{2} \|Aw_1 + e_1 b_1\|_2^2$$

and

$$\min_{w_2, b_2} \sum_{j=m_1+1}^{m_2} (x_j^T w_2 + b_2)^2 = \min_{w_2, b_2} \frac{1}{2} \|Bw_2 + e_2 b_2\|_2^2$$

where, $e_i \in R^{m_i}$ represent a vector of ones. The previous QPPs define the Least- Square problems corresponding to the linear systems, $Aw_1 + e_1b_1 = 0$ and $Bw_2 + e_2b_2 = 0$.

Since the zero vector belongs to the column spaces of both augmented matrices $\tilde{A} = [A \ e_1]$, $\tilde{B} = [B \ e_2]$ these systems always admit the trivial solution. Thus, without additional constraints, both QPPs collapse to the degenerated zero hyperplane.

Including the following constraint on both QPPs avert it from collapsing to the trivial solution while forcing each hyperplane to be as far as possible from the observations of the opposite class,

$$\begin{aligned} & \min_{w_1, b_1} \frac{1}{2} \|Aw_1 + e_1b_1\|_2^2 \\ & \text{s.t} \\ & \left\{ -(Bw_1 + e_2b_1) \geq e_2 \right. \\ & \text{and} \\ & \min_{w_2, b_2} \frac{1}{2} \|Bw_2 + e_2b_2\|_2^2 \\ & \text{s.t} \\ & \left\{ Aw_2 + e_1b_2 \geq e_1 \right. \end{aligned}$$

These constraints avert the hyperplanes from being collapsed into the zero hyperplane. Notice, now the trivial solution $w_i, b_i = 0$ would not be possible since it wont verify the constraints, $Aw_2 + e_1b_2 = 0 \not\geq e_1$ and $-(Bw_1 + e_2b_1) = -0 \not\geq e_2$. Moreover, the constraints forces the hyperplane to be at a distance of at least 1 from the observations of the opposite class.

For example, the constraint on the class 1 hyperplane, $-(Bw_1 + e_2b_1) \geq e_2$ impose $-(x_j^T w_1 + b_1) \geq 1 \ \forall x_j \in B$ hence, $x_j^T w_1 + b_1 \leq -1$ requiring class 2 observations to lie on the negative side of the hyperplane, being at least 1 unit away in functional margin.

Implicitly, it has been assumed that the classes are separable in the sense that there exists two non- parallel hyperplanes satisfying the separation constraints, $-(Bw_1 + e_2b_1) \geq e_2$ and $Aw_2 + e_1b_2 \geq e_1$. However, perfect separability rarely holds in practice hence, the previous hard-margin formulation must be relaxed, allowing class overlapping with some observations violating the separation constraints.

Thus, to allow violations of the separation constraints non-neagative slack variables are introduced, $\{\xi_1\}_{i=1}^{m_1}, \{\xi_2\}_{i=m_1+1}^{m_2}$ such that, $\xi_1 \geq 0, \xi_2 \geq 0$. Hence, the constraints are modified as follows,

$$-(Bw_1 + e_2b_1) \geq e_2 - \xi_2$$

and

$$(Aw_2 + e_1b_2) \geq e_1 - \xi_1$$

These slack variables relax the margin constraints imposed allowing some opposite class observations to be close to the hyperplane fitting the other class.

- $\xi_i = 0$, indicates the observation verifies the one unit separation from the other class hyperplane.
- $0 < \xi_i < 1$, indicates opposite class observations falling on the correct side but close to the other class hyperplane.
- $\xi_i = 1$ indicates opposite class observations falling exactly on the wrong hyperplane
- $\xi_i > 1$ indicates opposite class observations crossing the wrong hyperplane.

The slack variables are incorporated into the objective function through linear penalty terms. The hyperparameters c_1, c_2 define the trade-off between the minimization of the least-squares fitting error to the corresponding class and penalization of separation constraints violations.

Finally, the standard TWSVM algorithm solves the following Quadratic Programming Problem (QPP),

$$\min_{w_1, b_1, \xi_2} \frac{1}{2} \|Aw_1 + e_1b_1\|^2 + c_1 e_2^T \xi_2 \quad (1)$$

s.t

$$\begin{cases} -(Bw_1 + e_2b_1) + \xi_2 \geq e_2 \\ \xi_2 \geq 0 \end{cases}$$

and

$$\min_{w_2, b_2, \xi_1} \frac{1}{2} \|Bw_2 + e_2b_2\|^2 + c_2 e_1^T \xi_1 \quad (2)$$

s.t

$$\begin{cases} Aw_2 + e_1b_2 + \xi_1 \geq e_1 \\ \xi_1 \geq 0 \end{cases}$$

As stated above, opposite to classical SVM formulation, which maximizes the functional margin between classes, TWSVM minimizes the squared residuals between each observations of a given class and their associated hyperplane. Consequently, TWSVM replaces the functional margin maximization principle of standard SVM with a proximal least-squares fitting strategy, resulting in class-specific observations clustering around each hyperplane.

TWSVM with Individual Feature Selection

This work aims to implementate a TWSVM algorithm embeded with an individual feature selection per hyperplane as in (Bai, L. Wang, Z. Shao Y.H & Deng, N. Y (2014)). The further proposed embedded feature selection method extends the L_1 -TWSVM model that minimizes the L_1 -norm of the feature weight vectors. As it is well known, introducing the L_1 -norm ensures the L_1 -TWSVM model yields sparser feature weight vectors as a result while simplifying the learning process. The L_1 -TWSVM model is similarly to the earlier studied L_2 -TWSVM defines two QPP as follows,

$$\begin{aligned}
 & \min_{w_1, b_1, \xi_2} \|w_1\|_1 + c_1 \|Aw_1 + e_1 b_1\|_1 + c_3 e_2^T \xi_2 \\
 & \text{s.t} \\
 & \begin{cases} -(Bw_1 + e_2 b_1) \geq e_2 - \xi_2 \\ \xi_2 \geq 0 \end{cases} \\
 & \text{and} \\
 & \min_{w_2, b_2, \xi_1} \|w_2\|_1 + c_2 \|Bw_2 + e_2 b_2\|_1 + c_4 e_1^T \xi_1 \\
 & \text{s.t} \\
 & \begin{cases} (Aw_2 + e_1 b_2) \geq e_1 - \xi_1 \\ \xi_1 \geq 0 \end{cases}
 \end{aligned}$$

Notice, this L_1 -TWSVM could directly be used as an embedded feature selection method itself, such that whenever the two weights of a feature in the weight vectors w_1, w_2 are zeros the corresponding feature should be deleted.

The earlier representation of the L_1 -TWSVM compose a Nondifferentiable Programming Problem due to the presence on the objective function of non linear term as, $\|w_1\|_1 = \sum_{n=1}^N |w_n^{(1)}|$ and $\|Aw_1 + e_1 b_1\|_1 = \sum_{i=1}^{m_1} |\sum_{n=1}^N a_{i,n} w_n^{(1)} + b_1|$ in the definition of the first TWSVM hyperplane.

This problem can be transformed into a Differentiable Linear Programming Problem (DLPP) to be solved by the simplex algorithm variant implementated within the Gurobi optimizer just introducing the following auxiliar variables, $w_i = p_i - q_i$ as well as, $Aw_1 + e_i b_1 = s_1 - t_i$ and $Bw_2 + e_2 b_2 = s_2 - t_2$.

The auxiliary variables introduce a positive-negative decomposition of the weight vectors and residual terms, where each variable is expressed as the difference of two nonnegative variables. This allows their absolute values to be represented as the sum of these components, thereby enabling a linear formulation of the L_1 -TWSVM.

$$\begin{aligned}
& \min_{p_1, q_1, s_1, t_1, b_1, \xi_2} e_1^T(p_1 + q_1) + c_1 e_1^T(s_1 + t_1) + c_3 e_1^T \xi_1 \\
& \text{s.t} \\
& \begin{cases} A(p_1 - q_1) + e_1 b_1 = s_1 - t_1 \\ -B(p_1 - q_1) + e_2 b_1 \geq e_2 - \xi_2 \\ p_1, q_1, s_1, t_1, \xi_2 \geq 0 \end{cases} \\
& \text{and} \\
& \min_{p_2, q_2, s_2, t_2, b_2, \xi_1} e_2^T(p_2 + q_2) + c_2 e_2^T(s_2 + t_2) + c_4 e_2^T \xi_1 \\
& \text{s.t} \\
& \begin{cases} B(p_2 - q_2) + e_2 b_2 = s_2 - t_2 \\ A(p_2 - q_2) + e_1 b_2 \geq e_1 - \xi_1 \\ p_2, q_2, s_2, t_2, \xi_1 \end{cases}
\end{aligned}$$

Noticing, $s_1 = A(p_1 - q_1) + e_1 b_1 + t_1$ and $s_2 = B(p_2 - q_2) + e_2 b_2 + t_2$ the final DLPPs to be solved by Gurobi's Simplex algorithm implementation are defined,

$$\begin{aligned}
& \min_{p_1, q_1, t_1, b_1, \xi_2} e_1^T(p_1 + q_1) + c_1 e_1^T(A(p_1 - q_1) + 2t_1) + c_1 m_1 b_1 + c_3 e_1^T \xi_2 \\
& \text{s.t} \\
& \begin{cases} A(p_1 - q_1) + e_1 b_1 + t_1 \geq 0 \\ -B(p_1 - q_1) + e_2 b_1 \geq e_2 - \xi_2 \\ p_1, q_1, t_1, \xi_2 \geq 0 \end{cases} \\
& \text{and} \\
& \min_{p_2, q_2, t_2, b_2, \xi_1} e_2^T(p_2 + q_2) + c_2 e_2^T(B(p_2 - q_2) + 2t_2) + c_2 m_2 b_2 + c_4 e_2^T \xi_1 \\
& \text{s.t} \\
& \begin{cases} B(p_2 - q_2) + e_2 b_2 + t_2 \geq 0 \\ A(p_2 - q_2) + e_1 b_2 \geq e_1 - \xi_1 \\ p_2, q_2, t_2, \xi_1 \geq 0 \end{cases}
\end{aligned}$$

where $m_1 = e_1^T e_1$ and $m_2 = e_2^T e_2$ denote the total training observations of each class respectively.

Probability model for FTWSVM (TWSVM with Feature Selection)

Once the FTWSVM model has constructed both hyperplanes, a positive hyperplane $w_1^T x + b_1$ and a negative hyperplane $w_2^T x + b_2$, a new sample $\tilde{x}$ is assigned to either class depending on its proximity to these two hyperplanes.

In TWSVM, a new sample $\tilde{x}$ is assigned to the positive (negative) class if its perpendicular distance to the positive (negative) class hyperplane is smaller, $\frac{|w_1^T \tilde{x} + b_1|}{\|w_1\|} < \frac{|w_2^T \tilde{x} + b_2|}{\|w_2\|}$ (or $\frac{|w_1^T \tilde{x} + b_1|}{\|w_1\|} > \frac{|w_2^T \tilde{x} + b_2|}{\|w_2\|}$).

Equivalently, this decision rule can be expressed as a ratio, $\frac{|w_1^T \tilde{x} + b_1|}{|w_2^T \tilde{x} + b_2|} \cdot \frac{\|w_2\|}{\|w_1\|} < 1$ (or $\frac{|w_1^T \tilde{x} + b_1|}{|w_2^T \tilde{x} + b_2|} \cdot \frac{\|w_2\|}{\|w_1\|} > 1$). Thus, classification is based on comparing the relative distances of the new sample $\tilde{x}$ to the two hyperplanes, assigning it to the class of the closer one.

Shao, Y. H., Deng, N. Y., Yang, Z. M., Chen, W. J., & Wang, Z. (2012) propose a continuous scoring function to approximate the posterior probability of a sample $\tilde{x}$ belonging to a class. This score is constructed based on the geometric position of $\tilde{x}$ relative to two auxiliary hyperplanes.

Instead of working directly with the signed distances,

$$d_1(\tilde{x}) = \frac{w_1^T \tilde{x} + b_1}{\|w_1\|}, \quad d_2(\tilde{x}) = \frac{w_2^T \tilde{x} + b_2}{\|w_2\|}$$

consider a simple change of variables by using the linear combinations,

$$d_1(\tilde{x}) - d_2(\tilde{x}), \quad d_1(\tilde{x}) + d_2(\tilde{x})$$

These new variables induce the two auxiliary hyperplanes,

$$\begin{cases} d_1(\tilde{x}) - d_2(\tilde{x}) = 0 \\ d_1(\tilde{x}) + d_2(\tilde{x}) = 0 \end{cases}$$

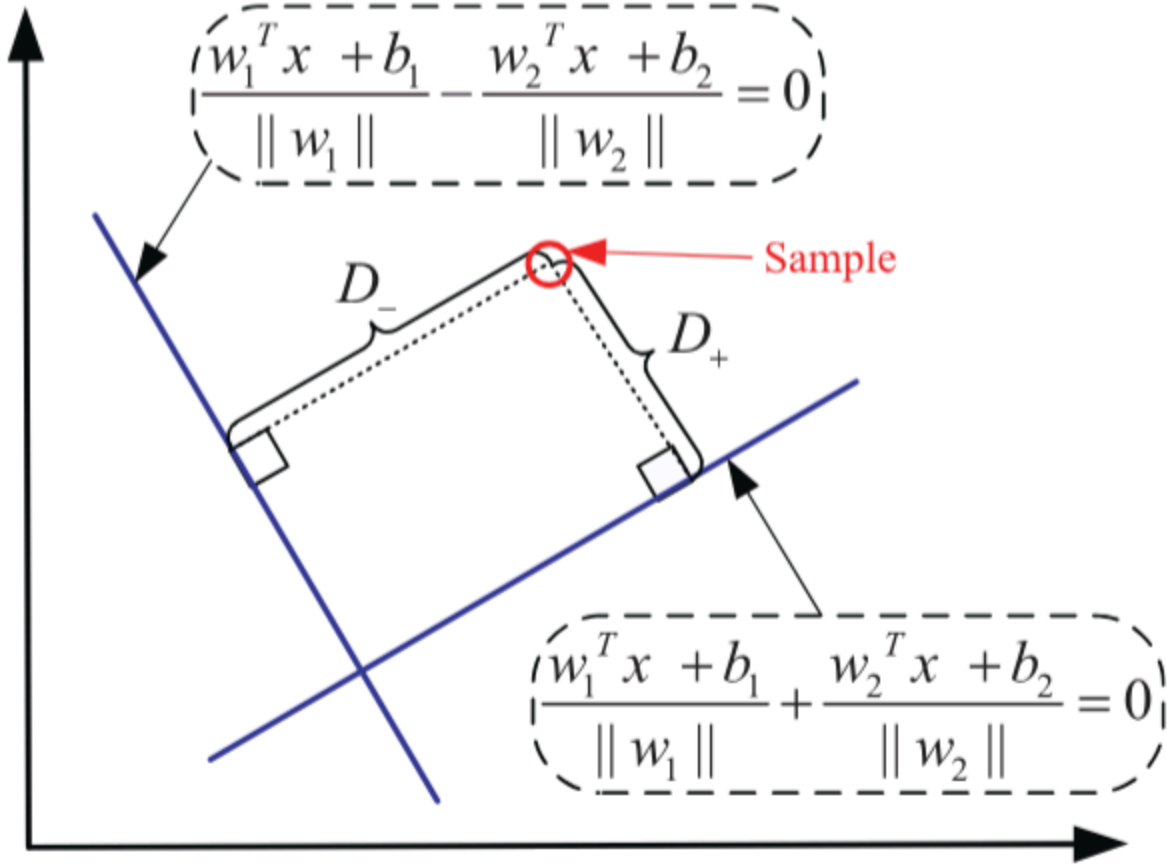
Notice, the first hyperplane enclose the region of samples that are equidistant from both TWSVM hyperplanes. Therefore, defining the classification boundary i.e the region of maximum uncertainty where, $P(y = +1) = \frac{1}{2}$. The second hyperplane introduces a symmetric geometric reference. It corresponds to configurations where the signed distances to the two decision hyperplanes have equal magnitude and opposite sign. It enables the normalization of distances, allowing the resulting score to be mapped into the interval $[0,1]$. These two auxiliary hyperplanes corresponds to the coordinate axes in the transformed space defined by the earlier stated change of variables.

$D_+(x)$ and $D_-(x)$ are defined as the distances from sample x to each auxiliary hyperplane,

$$D_+(x) = \frac{|d_1(x) + d_2(x)|}{\left\| \frac{w_1}{\|w_1\|} + \frac{w_2}{\|w_2\|} \right\|}$$

$$D_-(x) = \frac{|d_1(x) - d_2(x)|}{\left\| \frac{w_1}{\|w_1\|} - \frac{w_2}{\|w_2\|} \right\|}$$

Hence, one could consider two relevant quantities the minimum and maximum distances from a sample to a separating hyperplane as $d(x) = \min\{D_+(x), D_-(x)\}$ and $D(x) = \max\{D_+(x), D_-(x)\}$. So that we can define the ratio $\frac{d(x)}{D(x)}$.



The axis on the earlier figure represents the distances from a sample to the positive and negative decision boundaries defined by the TWSVM model, namely $d_1(x)$, $d_2(x)$. The figure is showing how a sample's position in the original distance space spanned by $(d_1(x), d_2(x))$ is mapped into the transformed coordinate system defined by $(d_1(x) - d_2(x), d_1(x) + d_2(x))$. Notice, the auxiliary hyperplanes $d_1(x) - d_2(x) = 0$ and $d_1(x) + d_2(x) = 0$ identified in the figure by blue continuous lines corresponds to the axis of the new transformed distance space. Moreover, the distances from the sample to each of the auxiliary hyperplanes, $D_+(x)$ and $D_-(x)$, are identified by dashed lines.

The previous figure provides the needed intuition that the greater the value of either $d(x)$ or $\frac{d(x)}{D(x)}$ for some sample, the more likely it belongs to the positive class. Hence, the proposed score function for capturing the degree of a sample belonging to the positive class was defined,

$$f(x) = \begin{cases} d(x) \left(\frac{d(x)}{D(x)} \right)^\alpha & \frac{|w_1^T \tilde{x} + b_1|}{\|w_1\|} < \frac{|w_2^T \tilde{x} + b_2|}{\|w_2\|} \\ 0 & \frac{|w_1^T \tilde{x} + b_1|}{\|w_1\|} = \frac{|w_2^T \tilde{x} + b_2|}{\|w_2\|} \\ -d(x) \left(\frac{d(x)}{D(x)} \right)^\alpha & \frac{|w_1^T \tilde{x} + b_1|}{\|w_1\|} > \frac{|w_2^T \tilde{x} + b_2|}{\|w_2\|} \end{cases}$$

where $\alpha > 0$ is a weight parameter. Recall, the bigger the value of this function $f(x)$ the more likely the sample x belongs to the positive class. It is worthnoticing the function $f(x) \in (-\infty, \infty)$.

Finally, the probability model is constructed by using a sigmoid to transform the results yields by the earlier function into probabilities,

$$P(y = +1|f(x)) = \frac{1}{1 + e^{af(x)+b}}$$

$$p(y = -1|f(x)) = 1 - P(y = +1|f(x))$$

Setting the weight parameter $\alpha = 1$, the parameters a, b for the sigmoid can be obtained by minimizing the cross- entropy between predicted probabilities p_i and target probabilities t_i in the following convex optimization problem,

$$\min_{a,b} - \sum_{i=1}^m \left(t_i \log(p_i) + (1 - t_i) \log(1 - p_i) \right)$$

$$\text{s.t. } p_i = P(y = +1 \mid f(x_i)),$$

$$t_i = \begin{cases} \frac{m_1 + 1}{m_1 + 2}, & \text{if } y_i = +1, \\ \frac{1}{m_2 + 2}, & \text{if } y_i = -1. \end{cases}$$